

AMIT KUMAR

Jaipur, Rajasthan | 6204139468 | amiiitaimlengg@gmail.com | [linkedin.com/in/amit-kumar-188809319](https://www.linkedin.com/in/amit-kumar-188809319) | github.com/AmitKr-06

OBJECTIVE

Final-year B.Tech Computer Science student specializing in Machine Learning and Generative AI (LangChain/RAG), seeking a Machine Learning Engineer / Data Scientist internship or full-time role. Experienced in building end-to-end ML pipelines from data analysis through deployment, with hands-on exposure to LLM-based application development.

EDUCATION

Bachelor of Technology (B.Tech) — Computer Science & Engineering

Arya Institute of Engineering Technology & Management, Jaipur (affiliated to Rajasthan Technical University) | CGPA: 7.5/10 | Expected Graduation: 2027

TECHNICAL SKILLS

Programming: Python, SQL, C++

ML / Deep Learning: NumPy, Pandas, Scikit-learn, PyTorch, XGBoost, LightGBM

NLP: NLTK, spaCy, Gensim, TF-IDF, Transformer

Generative AI / LLM: LangChain, Retrieval-Augmented Generation (RAG) — basics

Databases & Cloud: MongoDB Atlas, AWS S3

Deployment & Tools: FastAPI, Docker, Git/GitHub, DVC, MLflow, DagsHub

PROJECTS

Vehicle Insurance Response Prediction — End-to-End MLOps Pipeline | *Python, Scikit-learn, MongoDB Atlas, AWS S3, Docker*

- Built a 6-stage end-to-end MLOps pipeline (ingestion, validation, transformation, training, evaluation, pusher) predicting whether a customer will respond positively to a vehicle insurance offer, pulling raw data directly from a MongoDB Atlas collection
- Engineered a schema-validation stage that checks incoming data against a config-driven config/schema.yaml contract, plus a transformation stage with categorical encoding, feature scaling, and SMOTEENN to correct class imbalance
- Trained a RandomForestClassifier reaching an F1 score of ~0.93 (Precision ~0.88, Recall ~0.99), with an automated evaluation stage that compares each new model against the current production model in an AWS S3 registry and only promotes improvements
- Containerized the pipeline with Docker and parameterized it via environment variables for MongoDB and AWS credentials, enabling reproducible runs across environments

GitHub: github.com/AmitKr-06/Vehicle-Insurance

IMDB Movie Rating Predictor | *Python, Scikit-learn, XGBoost, LightGBM, NLP, DVC, MLflow*

- Built a modular, reproducible movie-rating prediction pipeline (cookiecutter-data-science project layout) that processes IMDB metadata and text features into model-ready datasets
- Applied NLP preprocessing (NLTK, spaCy, Gensim, TF-IDF) to unstructured text fields and combined the resulting features with structured metadata for modeling
- Trained and benchmarked XGBoost and LightGBM models against Scikit-learn baselines to identify the best-performing rating predictor
- Tracked experiments, datasets, and model versions with DVC, MLflow, and DagsHub for full reproducibility across pipeline runs

GitHub: github.com/AmitKr-06/IMDB-movie-predictor

SMS Spam Classifier | *Python, Scikit-learn, TF-IDF, Streamlit*

- Built and deployed a spam-detection system on 5,574 SMS messages, handling a 6.5:1 class imbalance using stratified sampling and precision/recall/F1 evaluation instead of raw accuracy
- Engineered structural meta-features (message length, digit presence, URL presence) combined with TF-IDF vectors; identified digit presence as the strongest spam signal (94.8% of spam vs. 15.7% of ham messages)
- Validated model stability using 5-fold cross-validation, achieving a mean F1 score of 0.895 (std. 0.022)

GitHub: github.com/AmitKr-06/sms-spam-classifier

CERTIFICATIONS

- IBM Machine Learning Professional Certificate — Coursera (6 courses: EDA for ML, Supervised Learning: Regression, Supervised Learning: Classification, Unsupervised Learning, Deep Learning & Reinforcement Learning, ML Capstone) — Completed July 2026